Section A

For each question there are four possible answers, **A**, **B**, **C** and **D**. Choose the **one** you consider to be correct.

1 In the discovery of an element, **E**, scientists found that it consist of 3 isotopes, ^{140}**E**, ^{142}**E** and ^{144}**E** in the atomic ratio of 2: 3: 1. Calculate the relative atomic mass of element **E** to 1 decimal place.

A 140.5

B 141.7

C 142.0

D 143.1

2 To determine the formula of an unknown hydrocarbon, **Y**, a student exploded 20 cm^3 of **Y** in 150 cm^3 of oxygen. At the end of the reaction, an expansion of 20 cm^3 was recorded, which contracted by 80 cm^3 when cooled. The resultant residual gas was then passed through an alkali which resulted in a further contraction to 10 cm^3. What is the formula of the unknown hydrocarbon **Y**?

A C_4H_8

B C_4H_{10}

C C_5H_8

D C_5H_{10}

3 In a redox reaction, 0.635 g of pure copper, Cu, required 20.00 cm^3 of 1.00 mol dm^{-3} nitric acid, HNO_3, for complete reaction. If Cu^{2+} ions was produced at the end of the reaction, what is the oxidation number of nitrogen in the products formed?

A +1

B +2

C +3

D +4

4 An element **G** in Period 2 has a general valence electronic configuration of ns^2 np^3. What is the formula of the flouride that is most likely to be formed by element **G**?

A GF_2

B GF_3

C GF_4

D GF_5

5 *Use of the Data Booklet is relevant to this question.*

The table below shows statements that 2 students made about the Ca^{2+} and Fe^{2+} ions.

student	statement
Q	Both ions were obtained by removing the electrons from the same orbital.
R	Both ions have more protons than neutrons and more protons than electrons.

Which students are correct?

A Student **Q** only

B Student **R** only

C Both Student **Q** and **R**

D None of them

6 Iodic acid is a strong acid that exist as a white solid at room temperature. It has a chemical formula of HIO_3 and dissolves readily in water to give H^+ and IO_3^- ions. Which of the following statements about the ion IO_3^- is true?

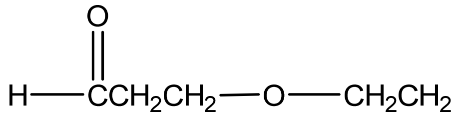
A It is a non-planar ion.

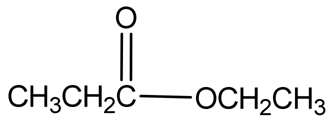
B It has a bond angle of 120°.

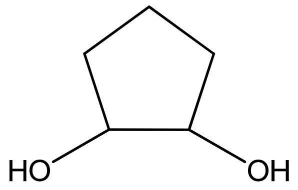
C Only sigma bonds are formed between iodine and oxygen in the ion.

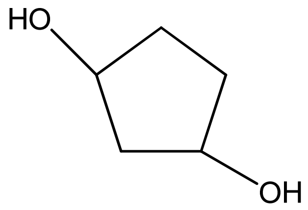
D The oxidation number of iodine in the ion is negative.

7 The following compounds are isomers with molecular formula $C_5H_{10}O_2$. Which of them is the least volatile?

A 

B 

C 

D 

8 Graphite is a common form of carbon that has many uses. One of them is as lubricants in machine parts. Which of the following properties of graphite allow for it to be used as lubricants?

A Each carbon atom in graphite is sp^2 hybridised.

B Graphite has a very high melting point.

C The interactions between layers of graphite is van der Waals forces of attraction.

D Graphite has an extended π-electron cloud above and below each plane of carbon atoms.

9 Which of the following arranges the ionic salts in increasing magnitude of their lattice energy?

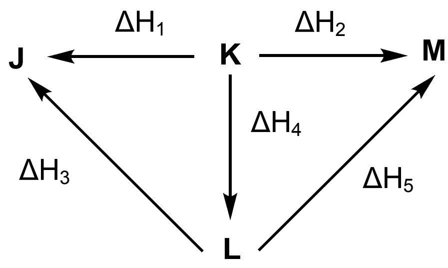
A $Al_2O_3 < MgO < MgBr_2 < NaBr$

B $NaBr < MgO < MgBr_2 < Al_2O_3$

C $MgO < NaBr < Al_2O_3 < MgBr_2$

D $NaBr < MgBr_2 < MgO < Al_2O_3$

10 The energy cycle below shows the reaction pathways between Compounds **J** – **M**.



What is the enthalpy change for the following reaction?

Compound **J** $\longrightarrow$ Compound **M**

A $\Delta H_1 + \Delta H_2$

B $\Delta H_2 - \Delta H_3 - \Delta H_4$

C $\Delta H_3 + \Delta H_5$

D $\Delta H_4 - \Delta H_1 - \Delta H_5$

11 To study the kinetics of a reaction between **X**, **Y** and **Z**, a student conducted a series of experiments and the following results were obtained:

Experiment	Vol of **X** / cm^3	Vol of **Y** / cm^3	Vol of **Z** / cm^3	Vol of H_2O / cm^3	Relative rate
1	20	15	10	55	1
2	40	15	10	35	4
3	40	25	15	20	6
4	40	15	20	25	8

What is the rate equation for the reaction?

A Rate = $[\mathbf{X}]^2[\mathbf{Z}]$

B Rate = $[\mathbf{X}]^2[\mathbf{Y}]$

C Rate = $[\mathbf{X}][\mathbf{Y}][\mathbf{Z}]$

D Rate = $[\mathbf{X}]^2[\mathbf{Y}][\mathbf{Z}]$

12 A technician found the following information in a handbook about an equilibrium reaction that the factory carries out on an industrial scale.

Temperature / °C	**Kc**
250	1×10^{-5}
500	1×10^{-4}

Pressure / atm	**% Product**
5	33
10	47

Based on the above information, what can be said about the equilibrium reaction that the factory carries out?

A The forward reaction is endothermic and there are more gaseous molecules in the products than reactants.

B The forward reaction is exothermic and there are less gaseous molecules in the products than reactants.

C The forward reaction is endothermic and there are less gaseous molecules in the products than reactants.

D The forward reaction is exothermic and there are more gaseous molecules in the products than reactants.

13 Which of the following equilibrium equations have no units for their equilibrium constant, K_c?

A $H_2O(g) + C(s) \rightleftharpoons H_2(g) + CO(g)$

B $CH_3CH_2OH(l) + CH_3COOH(l) \rightleftharpoons CH_3COOCH_2CH_3(l) + H_2O(l)$

C $2NO_2(g) \rightleftharpoons N_2O_4(g)$

D $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

14 The equation for the dissociation of water is as follows:

$$2H_2O(l) \rightleftharpoons H_3O^+(aq) + OH^-(aq) \quad \Delta H = +57\ kJmol^{-1}$$

Which of the following statements about the above equilibrium is correct?

A As temperature increases, the pH of water also increases.

B As the volume of water increases, the pH of water also increases.

C The summation of pH and pOH is 14 at all temperatures.

D $[H^+]$ will always equal to $[OH^-]$ at all temperatures.

15 Given that both NaOH and KOH are strong mineral bases, calculate the resultant pH when 20 cm^3 of 0.300 $moldm^{-3}$ of NaOH is mixed with 30 cm^3 of 0.200 $moldm^{-3}$ of KOH.

A 11.8

B 12.1

C 13.4

D 13.7

16 Which of the following properties show a decreasing trend across the elements of Period 3?

A pH of their chlorides in water

B Melting point of their oxides

C Electrical conductivity

D Ionic radius

17 An element in Period 3 has the following properties:

- It does not conduct electricity.
- Its first ionisation energy is larger than both the elements before and after it.

Which of the following statements about this element is correct?

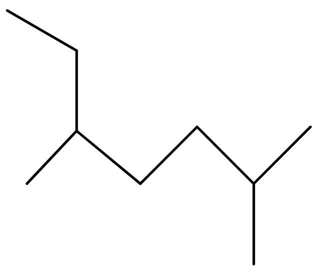
A It has a high melting and boiling point.

B It forms chlorides that are mildly acidic.

C It forms oxides that are very acidic.

D Its oxidation state in compounds usually follows the group number it belongs to in the Periodic Table.

18 What is the IUPAC name of the skeletal structure below?



A 2-ethyl-5,5-dimethylpentane

B 2-ethyl-5-methylhexane

C 1,2,5-trimethylhexane

D 2,5-dimethylheptane

19 How many positional isomers are there for a dichlorobutane?

A 5

B 6

C 7

D 8

20 Ethene is shaken with an aqueous mixture of iodine and sodium bromide.

Which of the following statements about the above reaction is ***incorrect***?

A There will at least one bromine atom found in all of the products formed.

B There will be a mixture of three different products formed.

C This is an addition reaction.

D The role of iodine in this reaction is an electrophile.

21 Which of the following properties of benzene is ***not*** associated with the ring of delocalised electrons?

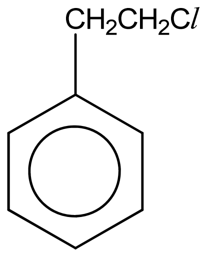
A Benzene tend to attract electrophiles rather than nucleophiles.

B Benzene tend to undergo substitution reactions rather than addition reactions

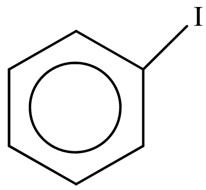
C Conversion of benzene to cyclohexane requires very harsh conditions.

D Benzene is a planar molecule with all the carbon atoms lying on the same plane.

22 The following are three different types of halogen derivatives:



1-chloro-2-phenylethane



iodobenzene

$CH_3CH_2CH_2CH_2CH_2CH_2Br$

1-bromohexane

A student then added ethanolic $AgNO_3$ separately to the 3 halogen derivatives above. Which of the following shows the correct order in which the precipitate appears, from the fastest to the slowest?

A 1-bromohexane, iodobenzene, 1-chloro-2-phenylethane

B 1-bromohexane, 1-chloro-2-phenylethane, iodobenzene

C iodobenzene, 1-bromohexane, 1-chloro-2-phenylethane

D 1-chloro-2-phenylethane, 1-bromohexane, iodobenzene

23 Which of the following reagents and conditions, when added to ethanol, involve the cleavage of the O-H bond in ethanol?

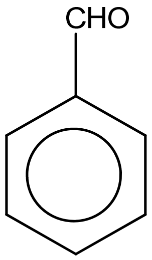
A CH_3COOH, excess concentrated H_2SO_4 and reflux

B I_2 dissolved in NaOH and warm

C PCl_5 at room temperature

D $K_2Cr_2O_7$, H_2SO_4(aq) and heat

24 Which of the following reagents can be used to distinguish between the following 2 compounds?



$CH_3CH_2CH_2C(=O)CH_2CH_2CH_3$

A 2,4-dinitrophenylhydrazine

B Fehling's solution

C Tollen's reagent

D $LiAlH_4$ in dry ether

25 Below are 3 compounds of carboxylic acids and its derivatives.

$CH_3CH(Cl)CH_2COOH$ $CH_3CH_2CH(Cl)COOH$ $CH_3CH_2CH_2COCl$

Which of the following arranges the pH of their solutions in increasing order when they are placed in water?

A $CH_3CH(Cl)CH_2COOH$, $CH_3CH_2CH(Cl)COOH$, $CH_3CH_2CH_2COCl$

B $CH_3CH_2CH_2COCl$, $CH_3CH(Cl)CH_2COOH$, $CH_3CH_2CH(Cl)COOH$

C $CH_3CH_2CH_2COCl$, $CH_3CH_2CH(Cl)COOH$, $CH_3CH(Cl)CH_2COOH$

D $CH_3CH(Cl)CH_2COOH$, $CH_3CH_2CH_2COCl$, $CH_3CH_2CH(Cl)COOH$

Section B

For each of the questions in this section, one or more of the thress numbered statements **1** to **3** may be correct.

Decide whether each of the statements is or is not correct (you may find it helpful to put a tick against the statements that you consider to be correct).

The responses **A** to **D** should be selected on the basis of

A	**B**	**C**	**D**
1, **2** and **3** are correct	**1** and **2** only are correct	**2** and **3** only are correct	**1** only is correct

26 $BeCl_2$ is able to react with PCl_3 to form a stable product.

Which of the statements about this reaction are correct?

1 The mole ratio of $BeCl_2 : PCl_3$ for this reaction is 1 : 2.

2 The product in this reaction is able to form because beryllium has energetically accessible orbitals to expand its octet configuration.

3 The bond angle about phosphorus does not change during this reaction.

27 A student investigated a reaction between compounds **T**, **U** and **V** and found that the order of reaction is first order with respect to **T**, zero order with respect to **U** and second order with respect to **V**. When a graph was drawn, a straight line was obtained.

Which of the following could be the axis of this graph?

1 Rate against [**T**]

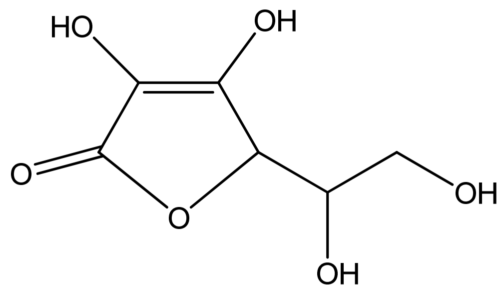
2 Rate against [**U**]

3 Rate against [**V**]^2

The responses **A** to **D** should be selected on the basis of

A	**B**	**C**	**D**
1, **2** and **3** are correct	**1** and **2** only are correct	**2** and **3** only are correct	**1** only is correct

28 Vitamin C is a common compound found in many citrus fruits such as oranges and pineapples. It has the following structure:



Which of the following statements about Vitamin C is ***incorrect***?

1 It is able to exhibit geometric isomerism.

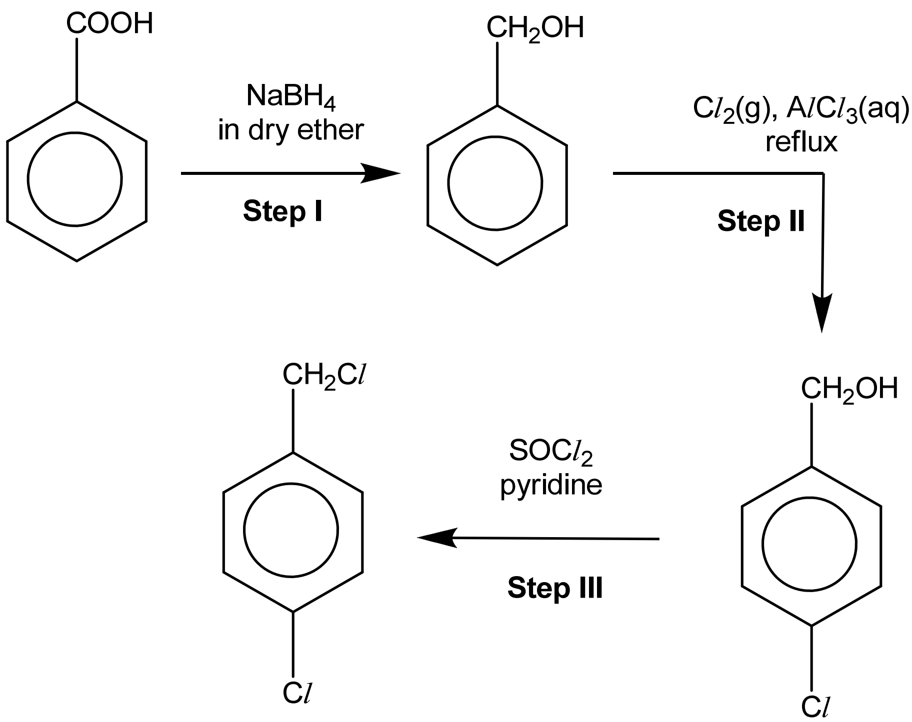
2 It will give orange crystals with Brady's reagent.

3 1 mole of Vitamin C will react with excess sodium to give 2 moles of hydrogen gas.

The responses **A** to **D** should be selected on the basis of

A	**B**	**C**	**D**
1, **2** and **3** are correct	**1** and **2** only are correct	**2** and **3** only are correct	**1** only is correct

29 The following synthetic pathway was proposed by a student:



Which of the following steps are unable to give the desired product that the student proposed in this synthesis?

1 **Step I**

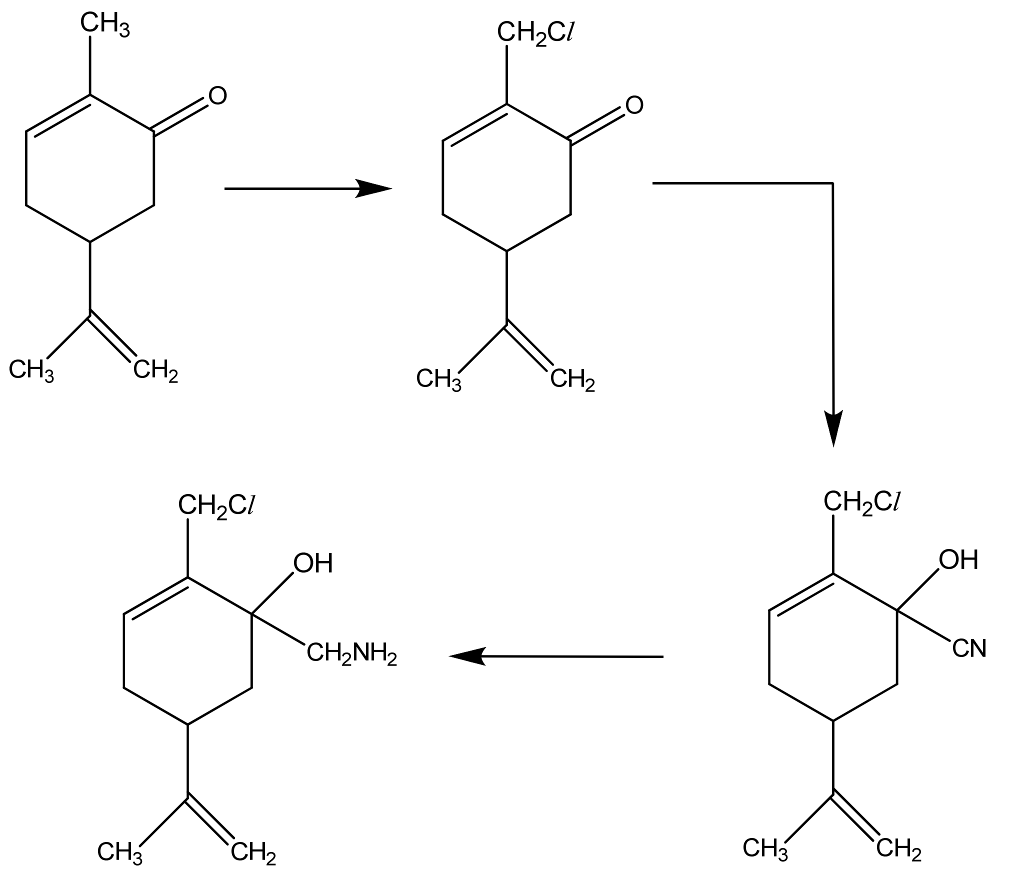
2 **Step II**

3 **Step III**

The responses **A** to **D** should be selected on the basis of

A	**B**	**C**	**D**
1, **2** and **3** are correct	**1** and **2** only are correct	**2** and **3** only are correct	**1** only is correct

30 Below is the synthesis pathway of Carvone, the main flavouring compound in spearmint oil.



Which of the following types of reactions are shown in the synthetic pathway above?

1 Addition

2 Reduction

3 Substitution

- THE END -

Answers:

1	**B**	6	**A**	11	**A**	16	**A**	21	**D**	26	**D**
2	**C**	7	**D**	12	**C**	17	**C**	22	**B**	27	**A**
3	**D**	8	**C**	13	**B**	18	**D**	23	**A**	28	**B**
4	**B**	9	**D**	14	**D**	19	**B**	24	**C**	29	**B**
5	**A**	10	**B**	15	**C**	20	**A**	25	**C**	30	**A**

[Turn over